

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.78)

Metric

$$ds^2 = \left(\frac{1}{3} \Lambda r^2 - 2H \right) du \otimes du - du \otimes dr - dr \otimes du + \frac{2r^2}{P^2} dx \otimes dx + \frac{2r^2}{P^2} dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Metric Functions

$$H(u, r, x, y) = H$$

$$P(u, x, y) = P$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(-\frac{1}{6} \Lambda r^2 + H \right) du + (1) dr$$

$$m = \left(\frac{r}{P} \right) dx + \left(\frac{i r}{P} \right) dy$$

$$\bar{m} = \left(\frac{r}{P} \right) dx + \left(-\frac{i r}{P} \right) dy$$

Contravariant components

$$l = (-1) \, dr$$

$$n = (-1) \, du + \left(-\frac{1}{6} \Lambda r^2 + H \right) \, dr$$

$$m = \left(\frac{P}{2r} \right) \, dx + \left(\frac{iP}{2r} \right) \, dy$$

$$\bar{m} = \left(\frac{P}{2r} \right) \, dx + \left(-\frac{iP}{2r} \right) \, dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u - \frac{1}{6} \Lambda r^2 + H \, \partial_r$$

$$\delta = \frac{P}{2r} \, \partial_x + \frac{iP}{2r} \, \partial_y$$

$$\bar{\delta} = \frac{P}{2r} \, \partial_x - \frac{iP}{2r} \, \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{1}{6}\Lambda r + \frac{H}{r} + \frac{\frac{\partial}{\partial u}P}{P}$$

$$\lambda = 0$$

$$\nu = \frac{P\frac{\partial}{\partial x}H}{2r} - \frac{iP\frac{\partial}{\partial y}H}{2r}$$

$$\alpha = \frac{\frac{\partial}{\partial x}P}{4r} - \frac{i\frac{\partial}{\partial y}P}{4r}$$

$$\beta = -\frac{\frac{\partial}{\partial x}P}{4r} - \frac{i\frac{\partial}{\partial y}P}{4r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{1}{6}\Lambda r - \frac{1}{2}\frac{\partial}{\partial r}H$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{\frac{\partial}{\partial x}P^2}{8r^2} + \frac{P\frac{\partial^2}{(\partial x)^2}P}{8r^2} - \frac{\frac{\partial}{\partial y}P^2}{8r^2} + \frac{P\frac{\partial^2}{(\partial y)^2}P}{8r^2} - \frac{H}{2r^2} - \frac{\frac{\partial}{\partial u}P}{2rP} + \frac{1}{4}\frac{\partial^2}{(\partial r)^2}H$$

$$\Phi_{12} = -\frac{P\frac{\partial^2}{\partial r\partial x}H}{4r} - \frac{iP\frac{\partial^2}{\partial r\partial y}H}{4r} - \frac{\frac{\partial^2}{\partial u\partial x}P}{4r} - \frac{i\frac{\partial^2}{\partial u\partial y}P}{4r} + \frac{\frac{\partial}{\partial u}P\frac{\partial}{\partial x}P}{4rP} + \frac{i\frac{\partial}{\partial u}P\frac{\partial}{\partial y}P}{4rP}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = -\frac{P\frac{\partial^2}{\partial r\partial x}H}{4r} + \frac{iP\frac{\partial^2}{\partial r\partial y}H}{4r} - \frac{\frac{\partial^2}{\partial u\partial x}P}{4r} + \frac{i\frac{\partial^2}{\partial u\partial y}P}{4r} + \frac{\frac{\partial}{\partial u}P\frac{\partial}{\partial x}P}{4rP} - \frac{i\frac{\partial}{\partial u}P\frac{\partial}{\partial y}P}{4rP}$$

$$\Phi_{22} = \frac{P^2\frac{\partial^2}{(\partial x)^2}H}{4r^2} + \frac{P^2\frac{\partial^2}{(\partial y)^2}H}{4r^2} + \frac{\frac{\partial}{\partial r}H\frac{\partial}{\partial u}P}{P} + \frac{\frac{\partial}{\partial u}H}{r} - \frac{2H\frac{\partial}{\partial u}P}{rP} - \frac{2\frac{\partial}{\partial u}P^2}{P^2} + \frac{\frac{\partial^2}{(\partial u)^2}P}{P}$$

$$\Lambda = \frac{1}{6}\Lambda - \frac{\frac{\partial}{\partial r}H}{3r} - \frac{\frac{\partial}{\partial x}P^2}{24r^2} + \frac{P\frac{\partial^2}{(\partial x)^2}P}{24r^2} - \frac{\frac{\partial}{\partial y}P^2}{24r^2} + \frac{P\frac{\partial^2}{(\partial y)^2}P}{24r^2} - \frac{H}{6r^2} - \frac{\frac{\partial}{\partial u}P}{2rP} - \frac{1}{12}\frac{\partial^2}{(\partial r)^2}H$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{\partial}{\partial r} \frac{H}{3r} + \frac{\partial}{\partial x} \frac{P^2}{12r^2} - \frac{P}{12r^2} \frac{\partial^2}{(\partial x)^2} P + \frac{\partial}{\partial y} \frac{P^2}{12r^2} - \frac{P}{12r^2} \frac{\partial^2}{(\partial y)^2} P + \frac{H}{3r^2} + \frac{1}{6} \frac{\partial^2}{(\partial r)^2} H$$

$$\Psi_3 = -\frac{P}{4r} \frac{\partial^2}{\partial r \partial x} H + \frac{iP}{4r} \frac{\partial^2}{\partial r \partial y} H + \frac{P}{2r^2} \frac{\partial}{\partial x} H - \frac{iP}{2r^2} \frac{\partial}{\partial y} H + \frac{\partial^2}{\partial u \partial x} P - \frac{i}{4r} \frac{\partial^2}{\partial u \partial y} P - \frac{\partial}{\partial u} P \frac{\partial}{\partial x} P + \frac{i}{4rP} \frac{\partial}{\partial u} P \frac{\partial}{\partial y} P$$

$$\Psi_4 = \frac{P^2}{4r^2} \frac{\partial^2}{(\partial x)^2} H - \frac{iP^2}{2r^2} \frac{\partial^2}{\partial x \partial y} H - \frac{P^2}{4r^2} \frac{\partial^2}{(\partial y)^2} H + \frac{P}{2r^2} \frac{\partial}{\partial x} H \frac{\partial}{\partial x} P - \frac{iP}{2r^2} \frac{\partial}{\partial y} H \frac{\partial}{\partial x} P - \frac{iP}{2r^2} \frac{\partial}{\partial x} H \frac{\partial}{\partial y} P - \frac{P}{2r^2} \frac{\partial}{\partial y} H \frac{\partial}{\partial y} P$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"